

SkillDzire Machine Learning

1. Python is a:

- A. Low-level language
- B. Machine language
- C. **High-level programming language**
- D. Assembly language

2. Python code blocks are defined using:

- A. Braces {}
- B. Semicolons ;
- C. **Indentation**
- D. Parentheses ()

3. Variable in Python is created when:

- A. Declared explicitly
- B. **A value is assigned**
- C. Program starts
- D. Compiler runs

4. Python follows:

- A. Static typing
- B. **Dynamic typing**
- C. Manual typing
- D. No typing

5. Which is a valid variable name?

- A. 1value
- B. value-1
- C. **value_1**
- D. value@1

6. Which of the following is a Python keyword?

- A. define
- B. **while**

- C. function
- D. loop

7. Which data type stores whole numbers?

- A. float
- B. **int**
- C. str
- D. bool

8. Which function checks the data type of a variable?

- A. datatype()
- B. check()
- C. **type()**
- D. isinstance()

9. Which keyword is used to check a condition?

- A. for
- B. while
- C. **if**
- D. switch

10. Which keyword is used for multiple conditions?

- A. if
- B. else
- C. **elif**
- D. when

11. Which block executes when all conditions are false?

- A. if
- B. elif
- C. **else**
- D. finally

12. Which loop is best when the number of iterations is known?

- A. while
- B. **for**

- C. do-while
- D. if

13. Which loop continues until the condition becomes false?

- A. for
- B. **while**
- C. if
- D. switch

14. Which statement exits a loop immediately?

- A. continue
- B. pass
- C. **break**
- D. return

15. Which statement skips the current iteration?

- A. break
- B. **continue**
- C. pass
- D. stop

16. Which keyword is used to define a function?

- A. function
- B. define
- C. **def**
- D. func

17. A function without return statement returns:

- A. 0
- B. False
- C. **None**
- D. Error

18. Which keyword is used to return a value from a function?

- A. output
- B. send

C. **return**

D. yield

19. Lambda functions are:

A. Named functions

B. Multi-line functions

C. **Anonymous functions**

D. Recursive functions

20. Lambda functions can contain:

A. Multiple statements

B. **Single expression**

C. No parameters

D. Only integers

21. map() function is used to:

A. Filter values

B. Reduce values

C. **Apply a function to all elements**

D. Sort data

22. filter() function is used to:

A. Modify values

B. **Select elements based on condition**

C. Sum elements

D. Sort data

23. reduce() function is used to:

A. Filter elements

B. Map values

C. **Combine elements into a single value**

D. Sort list

24. Which data structure is mutable?

A. Tuple

B. String

C. List

D. Frozen set

25. Index of first element in a list is:

A. 1

B. -1

C. 0

D. Any number

26. Which method adds an element to the end of a list?

A. insert()

B. add()

C. append()

D. extend()

27. Tuples are:

A. Mutable

B. Immutable

C. Dynamic

D. Editable

28. Sets do not allow:

A. Numbers

B. Strings

C. Duplicate elements

D. Objects

29. Dictionaries store data as:

A. Index-value pairs

B. Values only

C. Key-value pairs

D. Lists

30. Which symbol defines a dictionary?

A. []

B. {}

C. {}

D. <>

31. Which method returns all keys of a dictionary?

A. values()

B. items()

C. **keys()**

D. get()

32. Which operation updates a dictionary value?

A. append()

B. add()

C. **Assignment using key**

D. insert()

33. Which function is used to open a file?

A. file()

B. load()

C. **open()**

D. read()

34. Which mode opens a file for reading?

A. w

B. a

C. **r**

D. x

35. Which mode opens a file for writing?

A. r

B. **w**

C. a

D. rb

36. Which statement ensures file is closed automatically?

A. close

B. **finally**

C. **with**

D. pass

37. read() method is used to:

A. Write data

B. **Read entire file**

C. Close file

D. Delete file

38. write() method is used to:

A. Read data

B. **Write data to file**

C. Close file

D. Open file

39. Which module is used to handle CSV files?

A. pandas

B. file

C. **csv**

D. os

40. csv.reader() returns:

A. List

B. Dictionary

C. **Iterator**

D. Tuple

41. Python keywords are:

A. User-defined

B. Changeable

C. **Reserved words**

D. Variables

42. Which data type stores True or False?

A. int

B. str

C. **bool**

D. None

43. Which value represents absence of value in Python?

A. 0

B. False

C. Empty string

D. **None**

44. Which function converts string to integer?

A. str()

B. float()

C. **int()**

D. bool()

45. Python allows multiple assignment like:

A. a=1; b=2

B. **a, b = 1, 2**

C. a=b=1,2

D. assign a b

46. Which loop is best for unknown number of iterations?

A. for

B. **while**

C. if

D. switch

47. Which function returns number of elements in a list?

A. size()

B. count()

C. **len()**

D. length()

48. Which method removes the last element from a list?

A. remove()

B. **delete()**

C. **pop()**

D. **clear()**

49. Which operator checks equality?

A. =

B. ==

C. !=

D. <=

50. Python is popular because it is:

A. Complex

B. Hardware dependent

C. Low-level

D. **Simple, powerful, and versatile**

51. CSV stands for:

A. Comma Structured Values

B. Common Separated Values

C. **Comma Separated Values**

D. Column Separated Variables

52. Which Python module is used to read CSV files?

A. pandas

B. file

C. **csv**

D. os

53. csv.reader() returns:

A. List

B. Dictionary

C. **Iterator**

D. Tuple

54. csv.DictReader() returns each row as:

A. List

B. Tuple

C. Dictionary

D. Set

55. Which mode is used to read CSV files?

A. w

B. a

C. r

D. x

56. A simple CSV analyzer usually performs:

A. Sorting only

B. Reading, counting, and summarizing data

C. Model training

D. Visualization

57. Which statement ensures CSV file is closed automatically?

A. close()

B. finally

C. with

D. pass

58. Class in Python is a:

A. Variable

B. Function

C. Blueprint of objects

D. Package

59. Object is an instance of:

A. Module

B. Function

C. Class

D. Package

60. Constructor method in Python is:

A. init()

B. *init*

C. **init()**

D. constructor()

61. self keyword refers to:

A. Class

B. Module

C. **Current object**

D. Parent class

62. Methods defined inside a class are called:

A. Functions

B. **Member methods**

C. Packages

D. Variables

63. Best data structure to store contacts is:

A. List

B. Tuple

C. **Dictionary**

D. Set

64. In contact book, key usually represents:

A. Address

B. Phone number

C. **Contact name**

D. Index

65. CRUD operations stand for:

A. Copy Read Update Delete

B. **Create Read Update Delete**

C. Compile Run Update Debug

D. Create Remove Use Delete

66. Adding a contact in dictionary is an example of:

A. Read

B. **Create**

- C. Delete
- D. Update

67. Updating phone number is an example of:

- A. Create
- B. Read
- C. **Update**
- D. Delete

68. Using OOP in mini-project helps in:

- A. Slower execution
- B. **Better structure & reusability**
- C. More errors
- D. No benefit

69. SQL stands for:

- A. Simple Query Language
- B. **Structured Query Language**
- C. Sequential Query Language
- D. Server Query Language

70. Relational database stores data in:

- A. Files
- B. Objects
- C. **Tables**
- D. Graphs

71. Each row in a table is called:

- A. Column
- B. **Record**
- C. Field
- D. Schema

72. Each column represents:

- A. Table
- B. Row

C. Attribute

D. Index

73. SELECT is used to:

A. Insert data

B. Retrieve data

C. Delete data

D. Update data

74. WHERE clause is used to:

A. Sort data

B. Group data

C. Filter rows

D. Count rows

75. ORDER BY clause is used to:

A. Filter rows

B. Sort results

C. Join tables

D. Limit rows

76. LIMIT clause is used to:

A. Sort data

B. Filter columns

C. Restrict number of rows

D. Group rows

77. Default ORDER BY sorting is:

A. Descending

B. Ascending

C. Random

D. None

78. GROUP BY is used with:

A. WHERE

B. Aggregate functions

- C. JOIN
- D. LIMIT

79. COUNT() function returns:

- A. Sum
- B. Average
- C. **Number of rows**
- D. Maximum value

80. AVG() function calculates:

- A. Sum
- B. Count
- C. **Average**
- D. Maximum

81. SUM() function calculates:

- A. Count
- B. **Total value**
- C. Average
- D. Maximum

82. HAVING clause is used to:

- A. Filter rows
- B. **Filter grouped data**
- C. Sort data
- D. Limit rows

83. DISTINCT keyword removes:

- A. Null values
- B. **Duplicate rows**
- C. Sorted rows
- D. All rows

84. INNER JOIN returns:

- A. All rows
- B. **Matching rows from both tables**

- C. Only left table rows
- D. Only right table rows

85. LEFT JOIN returns:

- A. Matching rows only
- B. All rows from left table**
- C. All rows from right table
- D. No rows

86. RIGHT JOIN returns:

- A. Left table rows
- B. All rows from right table**
- C. Matching rows only
- D. No rows

87. FULL OUTER JOIN returns:

- A. Matching rows only
- B. Left table rows
- C. Right table rows
- D. All rows from both tables**

88. Subquery is:

- A. Query without SELECT
- B. Query inside another query**
- C. Join query
- D. Invalid query

89. Python connects to SQLite using:

- A. pymysql
- B. psycopg2
- C. sqlite3**
- D. sqlalchemy only

90. sqlite3.connect() returns:

- A. Cursor
- B. Connection object**

C. Database

D. Table

91. cursor.execute() is used to:

A. Fetch data

B. Execute SQL statements

C. Close DB

D. Commit data

92. commit() is used to:

A. Fetch rows

B. Save changes

C. Close DB

D. Rollback

93. Pandas reads SQL query using:

A. read_csv()

B. read_sql()

C. read_db()

D. sql_read()

94. SQLAlchemy is used for:

A. CSV handling

B. ORM & database abstraction

C. Visualization

D. Machine learning

95. Artificial Intelligence aims to:

A. Store data

B. Mimic human intelligence

C. Create databases

D. Write code

96. Machine Learning is a subset of:

A. Data Science

B. Artificial Intelligence

C. Deep Learning

D. Statistics

97. Deep Learning is a subset of:

A. AI

B. Data Science

C. Machine Learning

D. Databases

98. Machine Learning models learn from:

A. Rules only

B. Data

C. Hardware

D. IDE

99. Supervised learning uses:

A. Unlabeled data

B. Labeled data

C. No data

D. Random data

100. Unsupervised learning works with:

A. Labeled data

B. Unlabeled data

C. Output data

D. Target values

101. Classification is used to predict:

A. Continuous values

B. Categories

C. Clusters

D. Trends

102. Regression is used to predict:

A. Categories

B. Continuous values

- C. Labels
- D. Groups

103. Clustering is an example of:

- A. Supervised learning
- B. Unsupervised learning**
- C. Reinforcement learning
- D. Deep learning

104. Reinforcement learning is based on:

- A. Labels
- B. Clusters
- C. Reward and penalty**
- D. Rules

105. SQL primary key ensures:

- A. Sorting
- B. Uniqueness**
- C. Speed only
- D. Indexing only

106. Foreign key is used to:

- A. Sort data
- B. Link tables**
- C. Delete records
- D. Encrypt data

107. Best advantage of SQLAlchemy is:

- A. Speed only
- B. Database-independent code**
- C. CSV reading
- D. Visualization

108. Using Pandas with SQL helps in:

- A. Writing SQL only
- B. Data analysis & manipulation**

- C. Model training
- D. Deployment

109. ML model accuracy improves with:

- A. Less data
- B. **Quality data**
- C. Random data
- D. No preprocessing

110. AI, ML, and DL relationship is:

- A. $ML \subset DL \subset AI$
- B. **$DL \subset ML \subset AI$**
- C. $AI \subset ML \subset DL$
- D. Independent fields

111. csv.writer() is used to:

- A. Read CSV files
- B. **Write data to CSV files**
- C. Analyze CSV data
- D. Convert CSV to Excel

112. Which method reads CSV rows as dictionaries?

- A. reader()
- B. writer()
- C. **DictReader()**
- D. read_csv()

113. In CSV files, data is separated by default using:

- A. Tab
- B. Space
- C. **Comma**
- D. Semicolon

114. CSV analyzer can calculate average using:

- A. COUNT only
- B. **SUM and COUNT**

- C. JOIN
- D. WHERE

115. In OOP, bundling data and methods together is called:

- A. Inheritance
- B. Polymorphism
- C. **Encapsulation**
- D. Abstraction

116. Contact Book search operation is an example of:

- A. Create
- B. **Read**
- C. Update
- D. Delete

117. Deleting a contact from dictionary uses:

- A. remove()
- B. **del keyword**
- C. popitem() only
- D. erase()

118. Multiple contacts stored inside a class variable shows:

- A. Procedural programming
- B. **Object-oriented design**
- C. Functional programming
- D. Scripting

119. Which SQL statement inserts new records?

- A. SELECT
- B. UPDATE
- C. **INSERT**
- D. ALTER

120. Which SQL command modifies existing records?

- A. INSERT
- B. **SELECT**

C. UPDATE

D. DROP

121. Which SQL command removes records?

A. DROP

B. DELETE

C. REMOVE

D. TRUNCATE

122. ORDER BY salary DESC means:

A. Salary sorted ascending

B. Salary sorted descending

C. Salary grouped

D. Salary filtered

123. COUNT(*) counts:

A. Non-null values only

B. All rows

C. Distinct values only

D. Maximum value

124. AVG() ignores:

A. Zero values

B. NULL values

C. Duplicate values

D. Integers

125. JOIN condition is written using:

A. WHERE only

B. ON keyword

C. USING keyword only

D. SELECT

126. Which JOIN returns unmatched rows as NULL?

A. INNER JOIN

B. LEFT JOIN / RIGHT JOIN

C. CROSS JOIN

D. SELF JOIN

127. executemany() is used for:

A. Single row insert

B. Bulk insert/update

C. Fetch rows

D. Rollback

128. Parameterized queries protect against:

A. Deadlock

B. SQL Injection

C. Syntax error

D. Memory leak

129. Which placeholder is used in SQLite queries?

A. %s

B. :var

C. ?

D. {}

130. AI systems focus on:

A. Data storage

B. Intelligent decision making

C. File handling

D. Networking

131. ML algorithms improve performance by:

A. Hardcoding rules

B. Learning from data

C. Manual tuning only

D. Random guessing

132. Which is NOT a type of Machine Learning?

A. Supervised

B. Unsupervised

C. Reinforcement

D. **Procedural**

133. Spam detection is an example of:

A. Regression

B. **Classification**

C. Clustering

D. Dimensionality reduction

134. Contact Book mini-project best demonstrates:

A. File handling only

B. **OOP + Dictionary usage**

C. SQL only

D. ML only

135. Relational databases maintain data using:

A. Objects

B. Files

C. **Relationships between tables**

D. Graphs

136. SQLAlchemy ORM maps:

A. CSV to DB

B. Tables to lists

C. **Tables to Python classes**

D. Queries to files

137. Pandas read_sql() requires:

A. CSV file

B. **Database connection**

C. Excel file

D. Text file

138. ML model training requires:

A. Hardware only

B. **Data + algorithm**

- C. IDE only
- D. SQL only

139. Overfitting occurs when model:

- A. Performs poorly on training data
- B. Performs well on training but poorly on new data**
- C. Under-trains
- D. Has less data

140. Best practice before training ML model is:

- A. Deployment
- B. Data cleaning & preprocessing**
- C. Visualization only
- D. SQL joins

141. Which ML task predicts numeric value?

- A. Classification
- B. Clustering
- C. Regression**
- D. Reinforcement

142. K-Means is used for:

- A. Classification
- B. Clustering**
- C. Regression
- D. Prediction

143. AI > ML > DL means:

- A. DL is broader
- B. ML is independent
- C. DL is subset of ML, ML subset of AI**
- D. All are same

144. Contact Book update feature requires:

- A. New dictionary
- B. Key reassignment**

- C. Sorting
- D. Loop only

145. SQL LIMIT improves:

- A. Accuracy
- B. Performance & readability**
- C. Security
- D. Storage

146. ML classification output is:

- A. Continuous value
- B. Category/label**
- C. Cluster
- D. Sequence

147. Using Pandas + SQL is useful for:

- A. File creation
- B. Data analysis pipelines**
- C. IDE setup
- D. Game design

148. SQLite database is:

- A. Server-based
- B. Cloud-based
- C. File-based**
- D. In-memory only

149. AI-powered systems adapt by:

- A. Manual coding
- B. Learning from experience/data**
- C. Fixed rules
- D. Compilation

150. Mini-projects help learners by:

- A. Increasing theory
- B. Applying concepts practically**

- C. Reducing learning
- D. Avoiding coding

151. NumPy is mainly used for:

- A. Web development
- B. Numerical computing**
- C. File handling
- D. Visualization

152. NumPy array is:

- A. Heterogeneous
- B. Homogeneous**
- C. Immutable
- D. Dictionary-based

153. Which function creates an array of zeros?

- A. ones()
- B. empty()
- C. zeros()**
- D. array()

154. Vectorized operations in NumPy are:

- A. Slower
- B. Faster than loops**
- C. Same as loops
- D. Not supported

155. Pandas DataFrame is:

- A. 1D array
- B. 2D labeled data structure**
- C. Matrix only
- D. List

156. Which function reads CSV files in Pandas?

- A. readExcel()
- B. read_csv()**

C. open_csv()

D. load()

157. Which Pandas method shows summary statistics?

A. info()

B. **describe()**

C. summary()

D. stats()

158. Which method displays column data types?

A. type()

B. **dtypes**

C. info()

D. shape

159. Missing values in Pandas are represented by:

A. None only

B. 0

C. **NaN**

D. Blank

160. Which method fills missing values?

A. dropna()

B. **fillna()**

C. replace()

D. remove()

161. Duplicate rows are removed using:

A. unique()

B. **drop_duplicates()**

C. remove()

D. clear()

162. Outliers are extreme values that:

A. Improve accuracy

B. **Skew the distribution**

- C. Are duplicates
- D. Are missing

163. Feature engineering improves:

- A. Storage
- B. Model performance**
- C. Hardware
- D. Visualization

164. Matplotlib is used for:

- A. ML modeling
- B. Basic plotting**
- C. Data storage
- D. DL training

165. Seaborn is built on top of:

- A. NumPy
- B. Pandas
- C. Matplotlib**
- D. SciPy

166. Histogram is best for:

- A. Correlation
- B. Distribution analysis**
- C. Regression
- D. Classification

167. Boxplot is useful to detect:

- A. Mean
- B. Outliers**
- C. Frequency
- D. Variance

168. Mean is:

- A. Middle value
- B. Average**



- C. Most frequent value
- D. Spread

169. Median is:

- A. Average
- B. Middle value**
- C. Highest value
- D. Lowest value

170. Standard deviation measures:

- A. Central tendency
- B. Data spread**
- C. Mean
- D. Probability

171. Normal distribution is:

- A. Skewed
- B. Bell-shaped**
- C. Uniform
- D. Discrete

172. Bayes theorem is used to compute:

- A. Mean
- B. Conditional probability**
- C. Variance
- D. Correlation

173. Linear Regression predicts:

- A. Categories
- B. Continuous values**
- C. Clusters
- D. Sequences

174. Cost function used in Linear Regression is:

- A. Log loss
- B. Mean Squared Error**

C. Accuracy

D. Entropy

175. Overfitting means:

A. Poor training accuracy

B. Poor generalization

C. High bias

D. Under-training

176. L1 regularization is also called:

A. Ridge

B. Lasso

C. ElasticNet

D. Dropout

177. L2 regularization is also called:

A. Lasso

B. Ridge

C. ElasticNet

D. Bagging

178. Logistic Regression is used for:

A. Regression

B. Classification

C. Clustering

D. Dimensionality reduction

179. Logistic Regression uses which function?

A. Linear

B. ReLU

C. Sigmoid

D. Softmax

180. KNN classification is based on:

A. Tree depth

B. Distance between points

C. Probabilities

D. Gradients

181. Decision Trees split data based on:

A. Distance

B. Information gain / Gini index

C. Gradient descent

D. Random choice

182. Random Forest is an ensemble of:

A. Linear models

B. Decision Trees

C. SVMs

D. Neural networks

183. Confusion Matrix shows:

A. Features

B. TP, TN, FP, FN

C. Accuracy only

D. Loss

184. Precision measures:

A. All positives found

B. Correct positive predictions

C. Overall accuracy

D. Errors

185. Recall measures:

A. Precision

B. Ability to find all positives

C. Accuracy

D. Specificity

186. F1-score balances:

A. Accuracy & loss

B. Precision & recall

- C. Bias & variance
- D. Mean & median

187. ROC-AUC evaluates:

- A. Regression
- B. Binary classification performance**
- C. Clustering
- D. PCA

188. Unsupervised learning works with:

- A. Labeled data
- B. Unlabeled data**
- C. Target values
- D. Rewards

189. K-Means requires specifying:

- A. Distance
- B. Number of clusters**
- C. Learning rate
- D. Epochs

190. PCA is used for:

- A. Classification
- B. Dimensionality reduction**
- C. Clustering
- D. Regression

191. Hierarchical clustering output is:

- A. Tree model
- B. Dendrogram**
- C. Confusion matrix
- D. PCA plot

192. Apriori algorithm is used for:

- A. Regression
- B. Association rule mining**

C. Classification

D. Clustering

193. Cross-validation helps in:

A. Faster training

B. Better generalization evaluation

C. Feature creation

D. Scaling

194. GridSearchCV is used for:

A. Feature selection

B. Hyperparameter tuning

C. Data cleaning

D. Visualization

195. Feature scaling is important for:

A. Decision Trees

B. KNN & SVM

C. Random Forest

D. Naive Bayes

196. RFE stands for:

A. Recursive Feature Elimination

B. Recursive Feature Elimination

C. Random Feature Extraction

D. Reduced Feature Estimation

197. Pipelines help to:

A. Plot graphs

B. Automate preprocessing & modeling

C. Store data

D. Tune hardware

198. AdaBoost focuses on:

A. Random samples

B. Misclassified points

- C. Feature scaling
- D. Dimensionality reduction

199. Gradient Boosting builds models:

- A. Independently
- B. Sequentially**
- C. Randomly
- D. In parallel

200. XGBoost is known for:

- A. Simplicity
- B. Speed & performance**
- C. Visualization
- D. NLP only

201. Stacking combines:

- A. Features
- B. Predictions of multiple models**
- C. Datasets
- D. Algorithms only

202. Time series data is:

- A. Random
- B. Time-ordered data**
- C. Categorical
- D. Static

203. ARIMA stands for:

- A. Auto Random Integrated Moving Average
- B. AutoRegressive Integrated Moving Average**
- C. Advanced Regression Integrated Model
- D. Automatic Regressive Model

204. SARIMA is suitable for:

- A. Linear regression
- B. Seasonal time series**

C. Classification

D. Clustering

205. Moving average is used to:

A. Increase noise

B. Smooth time series

C. Create features

D. Detect outliers only

206. Trend refers to:

A. Noise

B. Long-term direction

C. Seasonality

D. Residual

207. Seasonality means:

A. Random variation

B. Repeating patterns over time

C. Trend only

D. Noise

208. MAPE stands for:

A. Mean Absolute Prediction Error

B. Mean Absolute Percentage Error

C. Maximum Absolute Percentage Error

D. Mean Accuracy Percentage Error

209. SMAPE improves MAPE by:

A. Ignoring zeros

B. Symmetric scaling

C. Squaring errors

D. Removing outliers

210. NumPy is faster because it is implemented in:

A. Python

B. C language

C. Java

D. R

211. Pandas is built on top of:

A. SciPy

B. NumPy

C. Matplotlib

D. Seaborn

212. Mean is sensitive to:

A. Median

B. Outliers

C. Mode

D. Distribution

213. Decision Trees are not sensitive to:

A. Noise

B. Feature scaling

C. Overfitting

D. Depth

214. Bias–Variance tradeoff refers to:

A. Accuracy vs speed

B. Underfitting vs overfitting

C. Precision vs recall

D. Mean vs median

215. Confusion matrix is applicable to:

A. Regression

B. Classification

C. Clustering

D. PCA

216. PCA components are:

A. Correlated

B. Orthogonal

- C. Random
- D. Categorical

217. K-Means minimizes:

- A. Classification error
- B. Within-cluster variance**
- C. Log loss
- D. Entropy

218. Elbow method helps to find:

- A. Accuracy
- B. Optimal number of clusters**
- C. Learning rate
- D. PCA components

219. Apriori uses:

- A. Distance
- B. Support & confidence**
- C. Gradient descent
- D. Entropy

220. Ensemble models generally:

- A. Reduce accuracy
- B. Improve robustness**
- C. Increase bias
- D. Overfit always

221. Gradient Boosting reduces:

- A. Variance only
- B. Bias gradually**
- C. Both equally
- D. Noise

222. Time series residuals represent:

- A. Trend
- B. Seasonality**

C. Unexplained noise

D. Average

223. Stationary time series has:

A. Changing mean

B. Constant mean & variance

C. Trend only

D. Seasonality only

224. Scaling is applied before PCA because:

A. PCA requires labels

B. PCA is variance-based

C. PCA needs normalization

D. PCA is linear

225. GridSearchCV evaluates models using:

A. Training set

B. Cross-validation

C. Test set only

D. Single split

226. Feature selection helps to:

A. Increase noise

B. Reduce overfitting

C. Increase dimensionality

D. Slow training

227. Random Forest reduces overfitting by:

A. Single tree

B. Averaging multiple trees

C. Feature scaling

D. PCA

228. XGBoost handles missing values by:

A. Removing rows

B. Learning optimal splits

- C. Mean imputation
- D. Dropping features

229. ROC curve plots:

- A. Accuracy vs loss
- B. TPR vs FPR**
- C. Precision vs recall
- D. Error vs epochs

230. AUC value closer to 1 indicates:

- A. Poor model
- B. Better classifier**
- C. Random guessing
- D. Overfitting

231. KNN performance degrades when:

- A. Low dimensions
- B. High dimensionality**
- C. Small data
- D. Scaling used

232. Curse of dimensionality affects:

- A. Linear Regression
- B. Distance-based algorithms**
- C. Trees
- D. Naive Bayes

233. Pipelines reduce:

- A. Model size
- B. Data leakage risk**
- C. Accuracy
- D. Features

234. Regularization strength is controlled by:

- A. Learning rate
- B. Alpha / lambda**

- C. Epochs
- D. Batch size

235. ElasticNet combines:

- A. Lasso only
- B. Ridge only
- C. **L1 and L2**
- D. Dropout

236. Classification threshold affects:

- A. Loss
- B. **Precision & recall**
- C. Accuracy only
- D. Features

237. PCA is unsupervised because:

- A. Uses labels
- B. **Ignores target variable**
- C. Needs tuning
- D. Reduces bias

238. Time series cross-validation must respect:

- A. Random sampling
- B. **Temporal order**
- C. Class balance
- D. Shuffling

239. Moving average window size controls:

- A. Trend
- B. **Smoothness**
- C. Seasonality
- D. Noise generation

240. MAPE fails when actual values are:

- A. Large
- B. **Zero**

- C. Negative
- D. Continuous

241. SMAPE addresses:

- A. Overfitting
- B. **MAPE asymmetry**
- C. Scaling
- D. Noise

242. Feature scaling is NOT required for:

- A. KNN
- B. SVM
- C. **Decision Trees**
- D. PCA

243. Random Forest feature importance is based on:

- A. Correlation
- B. **Impurity reduction**
- C. Distance
- D. Scaling

244. Cross-validation increases:

- A. Bias
- B. **Reliability of evaluation**
- C. Overfitting
- D. Noise

245. K-Means initialization problem is addressed by:

- A. Elbow method
- B. **k-means++**
- C. PCA
- D. Scaling

246. PCA reduces multicollinearity because components are:

- A. Correlated
- B. **Orthogonal**

- C. Random
- D. Discrete

247. Time series trend can be detected using:

- A. Histogram
- B. **Line plot**
- C. Boxplot
- D. Heatmap

248. Ensemble stacking uses:

- A. One model
- B. **Meta-learner**
- C. Distance metric
- D. PCA

249. Gradient Boosting can overfit if:

- A. Few trees
- B. **Too many trees**
- C. Low depth
- D. Early stopping used

250. Best end-to-end ML workflow is:

- A. Model → Data → Metrics
- B. Data → Model only
- C. **Data → Preprocess → Model → Evaluate → Optimize**
- D. Visualization → Deployment

Let's Learn and Grow